



of myths, and the magnitude of these myths multiplies when countries like India attempt to establish an ecosystem. One of the most prevalent yet most unaddressed myths around the semiconductor industry starts with the fake logic that fabs can only be successful in countries where there is a huge demand for semiconductors.

"It really is a myth. If you look at where the semiconductors are manufactured, whether it is fab or post-fab OSATs, they have got nothing to do with the market. The world's largest market for consumption of semiconductors is in the US. Yet, there is hardly any manufacturing in the US," shares Manickam.

The majority of semiconductor fabrication, as a matter of fact, takes place in countries like Taiwan and Korea. The back-end is carried out in countries like the Philippines, Malaysia, and Singapore. In 2020, the semiconductor industry in the United States, as per Statista, accounted for 47% of the global semiconductor market. However, taking a recent Wall Street Journal report into account, Taiwan is the country that is currently producing the most number of chips in the whole world. The distance between the top consumer (United States) and the top fabricator (Taiwan) is approximately 12,333km by sea. Additionally, the local market for semiconductor consumption in Taiwan stands nowhere close to the one in the States.

"You do not need to have a local market in order to play in this business. The narrative needs to change, and we need to convey that we are going to be global suppliers. It could be a small company, medium sized company, or a large company. You could be making a small frame that could go into a test equipment. You will probably make thousands of them," explains Manickam.

The second myth around semiconductors revolves around having a fab in the country to do anything backend around semiconductors. However, most forget that wafers are probably

the most convenient to get shipped across continents.

"You can literally have millions of dyes in a box, which can fit into an individual's hands. If you are working on a simple analogue device, you can have 20,000 to 25,000 chips from just one wafer. They are light and thin, making weight not an issue in the business," says Manickam.

Wafers that are fabricated in Taiwan are not necessarily packaged in Taiwan. Semiconductor fabrication companies in Taiwan ship a lot of wafers to the likes of Malaysia, Singapore, Philippines, and many other parts of the world for packaging and testing. These wafers then get assembled somewhere else.

"That's the second myth, we do not need a fab to be in the semiconductor business," explains Manickam.

The third myth, and probably the one that hinders most when it comes to setting up a semiconductor fabrication plant, revolves around the amount of investment required to do so. This myth, in some sense, is true. Setting up modern and up-to-date technology enabled semiconductor fabrication plants do require billions of dollars in investment. The likes of SK Hynix, TSMC, Samsung, and more announcing new fabrication plants worth billions of dollars do cement this fear, but not all the plants require the same kind of investment.

"It is probably true to some extent, but a lot depends on which segment you are going to play into. Whether you take the number of wafers that come out or you take it in terms of packaged parts, the majority of the units do not require 5, 7, or 3nm types of fabs. Majority is probably in the 28nm and above segment. There is plenty of business here," explains Manickam.

He adds, "Those fabs are not in the \$10 billion investment range. Depending on what you do, these could be in the half a billion to one billion investment bucket. Then you

can have specialised fabs like Silicon Carbide fabs. These are also now becoming mainstream, whether they are Silicon Carbide or GaN. The wafers are small, the kind of circuits that go in are different and simple. They are just in the \$20 to \$50 million investment range. The key to right investment lies in picking the right area and building your dominance into it."

There are thousands of small and medium companies in countries like Korea and Taiwan supporting the big names like TSMC, UMC, and ASC. A lot of these are family-run businesses. The key to their success, as Manickam shares, has been their vision to be global.

"The test tools that are required for testing semiconductors once they are packaged are more or less designed in India. But as these belong to multinational companies, they are taken away and sold at 10x the price. If we had that capability here in the country, it will go a long way," adds Manwani.

SMEs as India's strengths

While there are no brownie points for guessing that India missed the semiconductor boat a long time ago, there is definitely a whole cake waiting to be devoured for individuals and organisations who understand that the changing geo-political conditions have led us towards a new boat full of opportunities. However, following the same business model as other semiconductor OSAT and fabs might not work for India.

"We have some strengths in India. One of them is engineering, we are innovative, and we can scale the engineering as well. We can do a lot of customised work. If you look at all the other countries that are operating in the OSAT space, you will find that they have access to capital. They talk about 2 to 3 percent (interest on loan) whereas in India we talk about 10 to 12 percent. Cost of capital is very high for us," shares Manickam.

He adds, "They are who they are because they didn't have the kind of